Goldman Sachs Asset Management | IRS Operations & Settlements

Prompt Iteration Log

All 10 Prompts · 3 Iterations Each

AI Workflow Automation — IRS Trade Break Management | April 2026

Version 1 — Draft	Version 2 — Improved	Version 3 — Final

About This Log

This document records the full development history for each of the 10 AI prompts in the IRS Trade Break Automation Prompt Library. Every prompt went through three iterations. You can trace exactly what changed between each version and why.

An important note on how all prompts are framed: because an AI assistant does not have direct access to live trade systems, every prompt is written hypothetically — the analyst would paste or provide trade data as input text, and the AI responds based on that supplied information. No prompt assumes the AI can 'look at' or 'access' a real trade record directly.

The three versions follow a consistent pattern across all 10 prompts:

- Version 1 (red) — a simple, often vague first attempt. Shows what happens without proper prompt engineering.
- Version 2 (amber) — a meaningful improvement, but still with gaps that limit usefulness in a real workflow.
- Version 3 (green) — the production-ready version with full role conditioning, structured output, quantified parameters, and built-in oversight requirements.

Prompt 1 — Economic Field Mismatch Detection

Workflow	Given a hypothetical IRS trade record and a counterparty confirmation, identify which economic fields are mismatched and suggest resolution actions.

Version 1 — First Draft

Prompt Text:

"Imagine you have two records for the same IRS trade — one from an internal system and one from a counterparty. What fields might be different, and how would you identify them?"

What was wrong	What was added in V2
Too open-ended and conceptual. Asks about what 'might' differ rather than providing a structured task. No role, no output format, no resolution action — produces a general essay on trade breaks, not an actionable analysis.	Added analyst role, defined the input fields (notional, rate, dates), and asked for a list of differences rather than a discussion.

Version 2 — Improved

Prompt Text:

"You are a trade analyst. Suppose you are given two sets of IRS trade details — one from the internal booking system and one from the counterparty confirmation. Compare them field by field and list any differences you find."

What was still missing	What was added in V3

Role is too generic. Output format still unspecified — would produce narrative text. No instruction on which fields matter most for IRS economics. No resolution recommendation per field.	Upgraded role to IRS break analyst, defined the exact fields to compare, specified JSON output format, and added a resolution recommendation requirement per breaking field.

Version 3 — Final (Production-Ready)

Prompt Text:

"You are an IRS trade break analyst at a major investment bank. Suppose you are provided with two sets of trade details for the same IRS trade — one from the internal booking system and one from the counterparty confirmation — covering: trade ID, notional, fixed rate, floating rate index, effective date, maturity date, counterparty, and clearing status. Identify all economic fields that are mismatched. Output a structured JSON list with each breaking field, the internal value, the counterparty value, and a recommended resolution action."

Why this version works	Future improvement
Specific analyst role, full field list defined, structured JSON output enables pipeline integration, and resolution action per field adds direct operational value.	Could add product sub-type conditioning (vanilla IRS vs OIS vs basis swap) for more precise tolerance logic in future iterations.

Prompt 2 — Daily Break Prioritisation & Triage

Workflow	Given a hypothetical list of open trade breaks with their details, rank them by urgency using defined risk criteria.

Version 1 — First Draft

Prompt Text:

"Imagine you have a list of open trade breaks. How would you decide which ones to work on first?"

What was wrong	What was added in V2
Produces a discussion about prioritisation principles rather than a usable ranking tool. No criteria defined, no data inputs, no output format. Would give general advice that varies every run.	Introduced analyst role, named the input fields (break age, notional, counterparty), and asked for a ranked output.

Version 2 — Improved

Prompt Text:

"You are a settlements analyst. Suppose you have a report listing trade breaks with their age in days, notional size, and counterparty. Rank them from most to least urgent, considering how long they have been open and how large the notional is."

What was still missing	What was added in V3

Criteria partially defined but not quantified — 'how long' and 'how large' are ambiguous without thresholds. No mention of CCP-cleared trades which carry margin implications. No structured table output with per-break justification.	Added specific thresholds (>2 days, >$50M), CCP clearing as a distinct criterion, economic vs confirmation break category, and a ranked table output with justification column.

Version 3 — Final (Production-Ready)

Prompt Text:

"Act as a senior IRS settlements analyst. Suppose you are given a trade break report containing: trade ID, break age in days, notional size in USD, counterparty name, and break category (economic / confirmation / SSI). Rank the breaks from highest to lowest urgency using these criteria: (1) breaks outstanding more than 2 days, (2) notional above USD 50 million, (3) CCP-cleared trades, (4) economic breaks ranked above confirmation breaks. Return a ranked table with a justification column for each trade's priority score."

Why this version works	Future improvement
All four criteria are quantified and ordered. CCP-cleared trades are explicitly prioritised. Table output with justification makes the triage auditable and consistent across analysts.	Could add counterparty credit rating as a fifth criterion, and integrate directly with a live break feed via API.

Prompt 3 — Hard vs Soft Break Classification

Workflow	Given trade data from two sources, classify each field difference as either a genuine economic break (hard) or a tolerable rounding difference (soft).

Version 1 — First Draft

Prompt Text:

"If two trade records showed slightly different values for the same IRS trade, how would you decide if the difference is a real problem?"

What was wrong	What was added in V2
Completely conceptual — asks for a framework discussion rather than a classification task. No tolerance thresholds defined, no hard/soft output categories, no IRS product conventions referenced. Not usable in a workflow.	Changed to a classification task, introduced the concept of materiality vs rounding, and asked for amendment guidance.

Version 2 — Improved

Prompt Text:

"You are an IRS analyst. Imagine you have two records for the same trade with some small differences in values. Classify the differences as material (needing an amendment) or just rounding differences. Explain your reasoning."

What was still missing	What was added in V3
Better framing, but 'rounding' is not quantified. No IRS market conventions referenced (e.g., fixed rate to 4 decimal places, accrual days within ±1). No structured table output. Amendment advice is vague.	Added quantified tolerance examples specific to IRS conventions, defined hard/soft as formal output categories, added tabular output format, and included a column for whether an amendment is advisable even for soft breaks.

Version 3 — Final (Production-Ready)

Prompt Text:

"You are an expert in IRS product terms. Suppose you are given two sets of trade economics for the same IRS trade — Set A from the internal booking and Set B from the counterparty confirmation. Determine whether any differences fall within market-standard tolerance levels for IRS trades (for example: fixed rate rounded to 4 decimal places, accrual days within plus or minus 1). Flag any field outside tolerance as a 'hard break' requiring immediate resolution. Flag differences within tolerance as a 'soft break' and note whether an amendment is still advisable. Present results in a table."

Why this version works	Future improvement
Market-standard tolerance examples embedded as anchors. Hard/soft schema is clearly defined. The amendment advisability column addresses regulatory documentation needs even for non-material differences.	Tolerance thresholds could be moved to a separate config file so compliance can update them without modifying the prompt itself.

Prompt 4 — Counterparty Outreach Email Drafting

Workflow	Draft a professional, legally appropriate email to a counterparty to notify them of a hypothetical IRS trade break and request resolution.

Version 1 — First Draft

Prompt Text:

"Write an email to a counterparty about a trade break on one of our IRS trades."

What was wrong	What was added in V2
No trade variables, no tone guidance, no settlement timeline. Would produce a generic placeholder email. Risk of generating accusatory or vague language that could damage counterparty relationships or create legal exposure.	Introduced template variables for trade details, added a professional tone requirement, and asked the email to request action from the counterparty.

Version 2 — Improved

Prompt Text:

"Draft a professional email to [Counterparty Name] about a hypothetical trade break on IRS trade [Trade ID]. The break is on [Field Name] — our value is [Value A] and their value is [Value B]. Ask them to investigate and confirm their records."

What was still missing	What was added in V3
Variables in place — a clear improvement. However, no T+2 settlement convention reference for the resolution timeline. No explicit non-accusatory tone instruction. No structured break summary in the email body. Instructions to 'investigate' could come across as blaming.	Added T+2 convention reference, explicit non-accusatory tone requirement, structured break summary table in the email body, and four numbered content requirements to prevent omissions.

Version 3 — Final (Production-Ready)

Prompt Text:

"Act as a trade confirmation specialist. Draft a professional counterparty outreach email for a hypothetical IRS trade break with the following details: Trade ID [X], Counterparty [Y], Breaking Field [Z], Our Value [A], Their Value [B]. The email must: (1) state the discrepancy factually and without accusation, (2) ask the counterparty to review and confirm their records, (3) propose a resolution timeline aligned with T+2 settlement convention, (4) maintain a professional and cooperative tone throughout. Include a structured break summary table in the email body."

Why this version works	Future improvement
Role conditions for specialist communication style. Four numbered requirements prevent structural omissions. Non-accusatory tone protects the counterparty relationship. Break summary table enables the counterparty to action without further back-and-forth.	Could add counterparty tier sensitivity — a different tone for a G-SIB counterparty versus a smaller buy-side client.

Prompt 5 — Counterparty Break Pattern Analysis

Workflow	Analyse a hypothetical 30-day break history dataset to identify recurring counterparty or field-level patterns and recommend remediation actions.

Version 1 — First Draft

Prompt Text:

"Which counterparties cause the most trade break problems, and why?"

What was wrong	What was added in V2
Framing is adversarial — breaks are bilateral and root cause may sit with the bank, not the counterparty. No data input defined. No output structure. Would produce an opinion based on general knowledge, not actual data analysis.	Changed to a neutral analytical frame, introduced data as the source, and asked for counterparty frequency and field-level analysis.

Version 2 — Improved

Prompt Text:

"Analyse this hypothetical trade break dataset and tell me which counterparties have the most breaks and which fields break most often. Give me some recommendations."

What was still missing	What was added in V3
Neutral framing is a good improvement. However, analysis dimensions are underspecified — missing cleared vs bilateral split, time-to-resolution metric, and re-break rates. 'Some recommendations' is too vague. No counterparty-level action differentiation.	Enumerated four specific analysis dimensions, specified CSV as the data format, added time-to-resolution as a metric, and required per-counterparty remediation actions in the output.

Version 3 — Final (Production-Ready)

Prompt Text:

"Suppose you are given a hypothetical 30-day IRS break history dataset in CSV format covering the top 20 counterparties. Analyse it to identify: (1) which counterparties have the highest break frequency, (2) which economic fields break most often — such as fixed rate, notional, effective date, maturity date, or day count convention, (3) patterns by trade type — cleared versus bilateral, and (4) average time-to-resolution per break category. Summarise findings and provide a recommended remediation action for each counterparty trend identified."

Why this version works	Future improvement
Four analysis dimensions are fully specified. CSV format ensures reliable parsing. Per-counterparty remediation structure makes output directly actionable by the relationship management team.	Could add a month-on-month delta comparison to detect whether a counterparty's performance is improving or deteriorating over time.

Prompt 6 — End-of-Day Risk Summary for Management

Workflow	Generate a concise, standardised daily risk briefing for the Head of Settlements based on a hypothetical open breaks dataset.

Version 1 — First Draft

Prompt Text:

"Write a summary of open trade breaks for today."

What was wrong	What was added in V2
No audience defined — output would be too detailed and technical for senior management. No structure, no risk metrics, no call to action. Would produce a list of breaks rather than an executive briefing.	Specified the Head of Settlements as the audience, introduced total notional at risk as a metric, and added an age threshold flag.

Version 2 — Improved

Prompt Text:

"Write a summary of hypothetical open trade breaks for the Head of Settlements. Include the total notional at risk and flag any breaks that have been open for more than 3 days."

What was still missing	What was added in V3
Audience specified — a clear improvement. But no age bucket breakdown, no CCP margin implication flag, no recommended actions for the top risks, and no word limit to keep it executive-appropriate.	Added four-bucket age analysis (0-1, 1-2, 2-3, 3+ days), a CCP-cleared separate category, top-3 highest risk with recommended actions, and a 300-word maximum to enforce executive brevity.

Version 3 — Final (Production-Ready)

Prompt Text:

"You are an IRS operations risk assistant. Suppose you are given a hypothetical list of open trade breaks including their ages, notional amounts, and clearing status. Generate a daily risk summary suitable for the Head of Settlements including: (1) total USD notional at risk from unconfirmed trades, (2) number of breaks grouped by age — 0 to 1 day, 1 to 2 days, 2 to 3 days, and 3 or more days, (3) any breaks involving CCP-cleared trades that may attract margin implications, (4) the top 3 highest-risk breaks with a recommended immediate action for each. Format this as an executive briefing of no more than 300 words."

Why this version works	Future improvement
All four components are explicitly required. Four-bucket age analysis enables trend tracking over time. CCP flag is operationally critical. The 300-word cap makes the output usable in a management meeting without editing.	Could add a prior day comparison to show whether the overall break count and notional at risk is trending up or down.

Prompt 7 — Junior Analyst Investigation Guide

Workflow	Produce a step-by-step investigation guide for a Level 1 analyst handling a hypothetical complex IRS trade break.

Version 1 — First Draft

Prompt Text:

"How would a junior analyst go about resolving a trade break on an IRS trade?"

What was wrong	What was added in V2
No specific break context, no system references, no numbered steps. Would produce a high-level overview of settlements processes rather than a practical investigation guide. Pitched at the wrong level for a junior analyst.	Introduced a step-by-step format, specified the audience as junior, and named the key systems — Murex and MarkitWire.

Version 2 — Improved

Prompt Text:

"Walk a junior analyst through how to investigate a hypothetical IRS trade break step by step. Include steps for checking Murex and MarkitWire."

What was still missing	What was added in V3
Stepwise format and system references are improvements. However, no product knowledge context, no root cause hypothesis framework, Bloomberg VCON not included, and no escalation criteria for when the analyst should bring in a senior.	Added SME role conditioning, four root cause hypothesis types, Bloomberg alongside Murex and MarkitWire, and explicitly framed the output for a Level 1 analyst audience.

Version 3 — Final (Production-Ready)

Prompt Text:

"Act as an IRS product SME. Suppose a junior settlements analyst has escalated a hypothetical trade break to you involving specific breaking economic fields. Walk through step by step: (1) what each breaking economic field means in the context of an Interest Rate Swap, (2) the possible reasons it could be breaking — for example, a data entry error, a system mapping issue, a counterparty amendment, or a trade novation, (3) what specific investigations should be done in each system — Murex, MarkitWire, and Bloomberg — to resolve it. Present the guide as numbered steps written for a Level 1 analyst."

Why this version works	Future improvement
SME role produces domain-rich explanations. Four root cause types guide analyst thinking systematically. All three primary systems covered. Numbered steps with Level 1 framing ensures appropriate depth.	Could link each step to internal system navigation codes so the guide connects directly to in-house training materials.

Prompt 8 — Regulatory Obligation Screening

Workflow	Assess whether a hypothetical outstanding IRS trade break triggers EMIR, Dodd-Frank, or MAS reporting obligations and identify the escalation path.

Version 1 — First Draft

Prompt Text:

"Does an unresolved IRS trade break create any regulatory problems?"

What was wrong	What was added in V2
Completely undefined — no jurisdiction, no break characteristics, no specific rules named. Highly susceptible to the model inventing plausible-	Named two specific regulatory frameworks (EMIR and Dodd-Frank), introduced break age as a relevant input, and asked for obligation identification.

sounding but incorrect regulatory rule numbers. Not safe to use in any compliance context.	

Version 2 — Improved

Prompt Text:

"Suppose an IRS trade break has been outstanding for a hypothetical number of days. Identify whether this scenario triggers EMIR or Dodd-Frank reporting obligations and explain what should be done."

What was still missing	What was added in V3
Two jurisdictions named — a real improvement. Break age introduced as input. But counterparty type is missing (dealer vs buy-side changes the reporting obligation), no specific article references, no deadline calculation, no MAS jurisdiction, and no compliance officer escalation path.	Added counterparty type as an input variable, specific article references (EMIR Article 9, CFTC Part 45), deadline calculation, MAS as a third jurisdiction, escalation path output, and a compliance checklist output format.

Version 3 — Final (Production-Ready)

Prompt Text:

"You are a regulatory compliance assistant specialising in derivatives reporting. Suppose you are assessing a hypothetical IRS trade break scenario with the following details: the breaking fields, the counterparty type (dealer, buy-side, or CCP), the applicable jurisdiction (EMIR, Dodd-Frank, or MAS), and the number of days the break has been outstanding. Determine: (1) whether this scenario triggers any regulatory reporting obligation, (2) the specific rule reference — for example EMIR Article 9 or CFTC Part 45, (3) the deadline for resolution before a reportable event occurs, (4) the recommended escalation path within the firm. Output as a compliance checklist. Note: all outputs must be reviewed by a qualified compliance officer before any action is taken."

Why this version works	Future improvement
Three jurisdictions covered. Counterparty type correctly conditions the analysis. Specific article references ground outputs in verifiable sources. The compliance officer sign-off instruction is built into the prompt itself.	Should include a version date on the output so recipients know when the regulatory parameters were last reviewed by Legal.

Prompt 9 — Multi-System FpML / Murex Reconciliation

Workflow	Map and compare fields between a hypothetical MarkitWire FpML message and a Murex trade extract to produce a structured reconciliation report.

Version 1 — First Draft

Prompt Text:

"If you had two trade records for the same IRS trade from different systems, how would you compare them?"

What was wrong	What was added in V2
Asks for a methodology discussion rather than performing a reconciliation task. No format context — the model does not know one record is FpML XML and the other is a Murex extract with different field naming conventions. No output structure.	Changed to an action task, named both source formats (MarkitWire FpML and Murex), and asked for a field mapping and difference report.

Version 2 — Improved

Prompt Text:

"Suppose you have a hypothetical MarkitWire FpML message and a Murex extract for the same IRS trade. Map the equivalent fields between the two formats and highlight any differences in values."

What was still missing	What was added in V3
Format identification is a significant improvement. Semantic mapping introduced. However, no instruction to detect fields present in one source but absent from the other. No match status classification. No structured table with named columns.	Added absent-field detection, match/mismatch/absent status column, a notes column, and a five-column table format with explicitly named columns for downstream system integration.

Version 3 — Final (Production-Ready)

Prompt Text:

"You are a trade data analyst. Suppose you are given a hypothetical MarkitWire VCON message in FpML format and a Murex trade booking extract for the same IRS trade. Perform a field-by-field comparison by: (1) mapping equivalent fields between the two formats — for example, MarkitWire effectiveDate maps to Murex START_DATE, (2) identifying all value discrepancies, (3) flagging any fields present in one source but absent in the other, (4) producing a reconciliation report as a table with these columns: Field Name | MarkitWire Value | Murex Value | Match Status | Notes."

Why this version works	Future improvement
Real field name examples anchor the semantic mapping. Four-step task decomposition ensures completeness. Absent-field detection closes a critical gap from V2. The five-column table is directly importable into a reconciliation tool.	Could add an FpML schema version parameter to handle legacy trade formats and a condition for multi-leg swap structures.

Prompt 10 — Root Cause Analysis & Process Improvement

Workflow	Analyse a hypothetical month of break resolution log data to identify systemic issues and propose ranked process improvement initiatives.

Version 1 — First Draft

Prompt Text:

"Why do so many IRS trade breaks occur, and what can operations teams do about it?"

What was wrong

A general discussion question, not an analytical task. No data input, no output structure, no cost quantification, no improvement ranking. Would produce generic industry observations rather than firm-specific, data-driven recommendations.

What was added in V2

Changed to an analysis task using hypothetical log data, introduced root cause identification, and asked for improvement suggestions.

Version 2 — Improved

Prompt Text:

"Suppose you have a month of hypothetical trade break resolution logs. Analyse them to identify the top root causes and suggest some process improvements the operations team could make."

What was still missing

Data-driven framing is a real improvement. However, no cost quantification, no effort-vs-impact ranking for improvements, no executive summary output, and no governance committee framing. Suggestions would be generic rather than prioritised.

What was added in V3

Added operational cost quantification using analyst time as the unit, effort-vs-impact ranking for the three improvement initiatives, and an executive summary output format suitable for an Operations Review Committee.

Version 3 — Final (Production-Ready)

Prompt Text:

"You are a continuous improvement analyst in IRS operations. Suppose you are given a hypothetical month of break resolution log data including break types, root causes as coded by analysts, time-to-resolution, and re-break rates. Produce a root cause analysis report that: (1) identifies the top 3 systemic issues driving breaks, (2) quantifies the operational cost of each issue using average analyst time per break as the metric, (3) proposes 3 process improvement initiatives ranked by implementation effort versus expected break reduction, (4) includes a one-page executive summary suitable for presentation at an Operations Review Committee meeting."

Why this version works

All four components are well-defined. Cost quantification makes the business case tangible. Effort-vs-impact ranking enables investment prioritisation. The executive summary format makes the output directly usable in a governance forum.

Future improvement

Could add a year-on-year comparison dimension and, where available, benchmarks against industry break rate data to contextualise performance.

Key Lessons from the Iteration Process

Five consistent patterns emerged across all 10 prompts that separated a weak first draft from a production-ready final version:

- Role conditioning matters. 'You are an IRS trade break analyst at a major investment bank' produces far more targeted outputs than 'you are a trade analyst' or no role at all.
- Output format must be stated explicitly. JSON, table, numbered steps, executive briefing — without this, the model defaults to narrative prose that cannot feed into downstream systems.
- Quantified thresholds beat vague guidance. '>2 days' and '>USD 50 million' are operationally actionable. 'Old breaks' and 'large trades' are not.
- Real domain examples reduce hallucination risk. Embedding actual field names like EMIR Article 9, effectiveDate vs START_DATE, and Actual/360 grounds the model's output in verifiable reality rather than plausible-sounding invention.
- Hypothetical framing keeps prompts honest. Because the AI does not have access to live trade systems, every prompt uses 'suppose you are given...' language. This makes the prompt accurate and deployable — the analyst provides the data, the AI analyses it.
- Human oversight must be designed in, not assumed. Final versions explicitly name the review gate — for example, 'must be reviewed by a qualified compliance officer before any action is taken' — because in financial services, automation without oversight creates real regulatory and financial risk.

End of Iteration Log